PROJECT 02: Progressive Electricity Billing

1. Problem Statement

Calculate household electricity bills using progressive unit slabs, fixed charges, surcharge and rebate; compare monthly bills and flag unusually large increases.

2. Objectives

• Calculate energy charges by progressive slabs.
• Add fixed charges and surcharge, then apply eligible rebate.
• Compare consecutive monthly bills.
• Flag a bill increase greater than 30%.

3. Assumptions

• Illustrative slabs: first 100 units at Rs.1.50, next 100 at Rs.2.50, next 300 at Rs.4.00, remaining units at Rs.6.00.
• Fixed charge is Rs.50; surcharge is 5% of energy charge.
• Rebate is Rs.20 when consumption is 100 units or less.
• An increase above 30% over the previous month is unusual.

4. Algorithm

1. Read monthly unit consumption.
2. Reject negative unit values.
3. Apply each slab only to the units within that slab.
4. Add fixed charge and surcharge, and subtract any rebate.
5. Compare this bill with the previous bill.
6. Display bill amount and any unusual-increase warning.

5. Flowchart

6. Pseudocode

```
START
  FOR each month
    READ units
    IF units < 0: REPORT invalid input
    ELSE
      CALCULATE slab-wise energy charge
      ADD fixed charge and 5% surcharge
      IF units <= 100: SUBTRACT rebate
      CALCULATE total bill
      IF previous bill exists AND total > previous * 1.30
        FLAG unusual increase
      DISPLAY bill
      SET previous = total
END
```

7. Python Code

```
def energy_charge(units):
    slabs = [(100, 1.50), (100, 2.50), (300, 4.00)]
    remaining, charge = units, 0.0
    for limit, rate in slabs:
        used = min(remaining, limit)
```

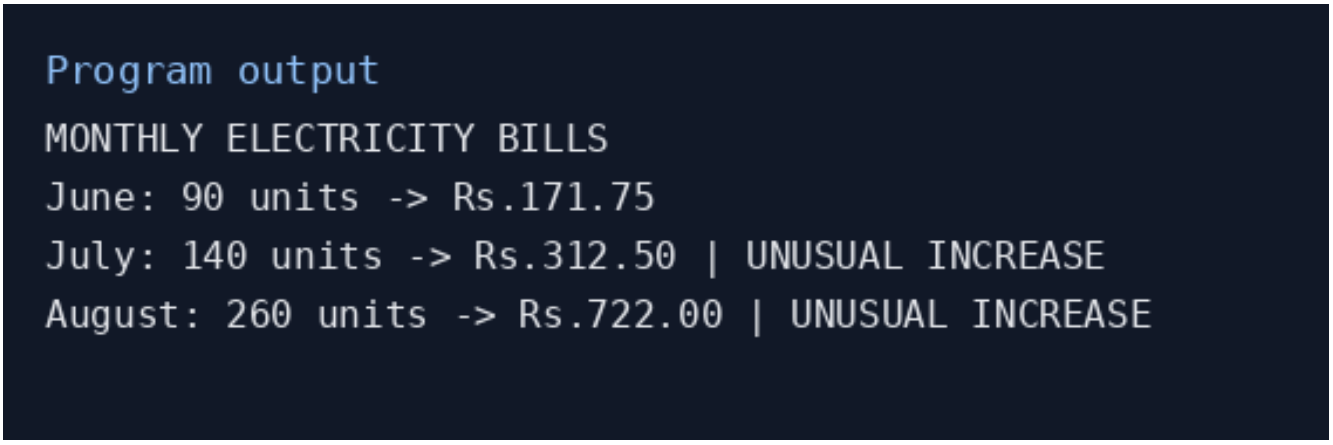
```
        charge += used * rate
        remaining -= used
        if remaining <= 0:
            break
    if remaining > 0:
        charge += remaining * 6.00
    return charge

def bill(units):
    if units < 0:
        raise ValueError("Units cannot be negative")
    energy = energy_charge(units)
    return energy + 50 + energy * 0.05 - (20 if units <= 100 else 0)

months = [("June", 90), ("July", 140), ("August", 260)]
previous = None
print("MONTHLY ELECTRICITY BILLS")
for month, units in months:
    total = bill(units)
    warning = " | UNUSUAL INCREASE" if previous is not None and total > previous * 1.30 else ""
    print(f"{month}: {units} units -> Rs.{total:.2f}{warning}")
    previous = total
```

8. Test Cases

Test	Input / condition	Expected result
Low consumption	90 units	Bill includes Rs.20 rebate
Slab boundary	100 units	First slab rate applies to all 100 units
Higher consumption	260 units	All applicable slabs are charged progressively
Invalid input	-10 units	Raises ValueError: Units cannot be negative

9. Output Screenshot

The following screenshot-style image was generated from the captured output of the program execution.



Captured output

```
MONTHLY ELECTRICITY BILLS
June: 90 units -> Rs.171.75
July: 140 units -> Rs.312.50 | UNUSUAL INCREASE
August: 260 units -> Rs.722.00 | UNUSUAL INCREASE
```

10. Conclusion

The program applies progressive pricing rather than one rate to all units and highlights monthly increases that cross the configured threshold.

Academic note: The program was executed with the sample data shown in the source code. Policies, rates, thresholds and exchange rates are illustrative and should be adjusted to the assignment's official requirements.